

Question 1

Objective

To determine if it is worthwhile to spend money to attempt to apply the buildable area and footprint fitting solution to Calgary.

Attempts

From Open Data:

- Parcel Address
 - o Only points, no polygons
- Parcel Address and lat/long
 - o Only points, no polygons
- Current Year Property Assessments (Parcel)
 - o Missing parcels not assessed in the current year
- Historical Property Assessments (Parcel)
 - o Up to 21 polygons to represent a single parcel, one for each assessed year
 - o Massive dataset, an error is received when attempting to download

Evidence

- Unlike Burnaby, Calgary does not provide a clean dataset of all parcels for free. Calgary makes this available for purchase, \$60 per section. There are about 320 sections, so the total cost would be about \$19,000.

<https://cityonline.calgary.ca/ownership-parcel-fabric-opf-by-section---shapefile/5637144619.p>

Diagnosis

We didn't know the clean parcel data for Calgary is only available by purchasing until looking into it more deeply.

Proposed Next Steps

We only buy one section for \$60 which should contain several hundred parcels. Alternatively, we could support an alternate secondary city instead of Calgary.

Decision Ask

Do you think \$60 is worth being able to have a proof-of-concept prototype for a GIS implementation for Calgary? Would you rather a different free city be used instead?

Question 2

Objective

To determine if we should consider ignoring the smallest N accessory structures when defining buildable area.

Attempts

The current buildable area solution has a parameter, IGNORE_N_SMALLEST_ACCESSORIES (N). The smallest N accessory structures are ignored when defining buildable area. If N is 0, then all existing structures are considered. Essentially, this give a way to expand buildable area by assuming the N smallest accessory structures can be demolished. The largest building on a parcel is labelled as “primary front”, all others are labelled as “accessory”.

Evidence

Buildable area script with IGNORE_N_SMALLEST_ACCESSORIES parameter:

https://github.com/ubco-mds-2025-labs/w2025-data599-capstone-projects-green-metrics-technology/blob/main/code/gis_files/burnaby/get_buildable_areas.py

Full QGIS implementation for Burnaby: https://github.com/ubco-mds-2025-labs/w2025-data599-capstone-projects-green-metrics-technology/tree/main/code/gis_files/burnaby



Diagnosis

We are unsure if it is appropriate to consider demolishing accessory structures when building a new laneway home.

Proposed Next Steps

Our GIS dashboard should only show buildable areas with all existing structures considered. However, after the project is complete, someone else could expand on the solution to make the IGNORE_N_SMALLEST_ACCESSORIES parameter accessible to the user.

Decision Ask

Is it appropriate to consider demolishing accessory structures when building a new laneway home?
Do you agree that we should consider all existing structures for our GIS dashboard?